Greenhouse Management Realization

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1 The Data Source

I have tried to find a greenhouse dataset that meets the requirements that I need such as: date&time, temperature, humidity, soil moisture, and the light available for photosynthesis. I have encountered several websites that give real data sets for greenhouses, but they do not satisfy all the requirements I need. Then I attended this data analysis workshop and was told that I can use a sample data set for my project to start using a website to generate a mock data set [1].

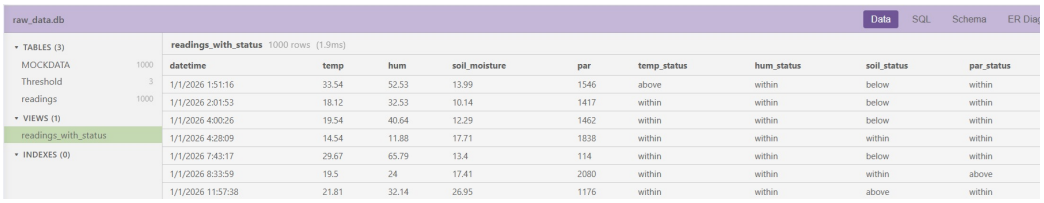
I have generated the data set as a csv file as:

Mock_Data.csv

I also have a threshold table [2] that I used for temperature and humidity, soil moisture [3], and PAR (Photosynthetically Active Radiation) [4] saved as:

Threshold.xls

I then worked on importing those tables into a database using sqlite3 in Python. I added another table named readings and created a view with reading status as shown in Figure 1.



datetime	temp	hum	soil_moisture	par	temp_status	hum_status	soil_status	par_status
1/1/2026 1:51:16	33.54	52.53	13.99	1546	above	within	below	within
1/1/2026 2:01:53	18.12	32.53	10.14	1417	within	within	below	within
1/1/2026 4:00:26	19.54	40.64	12.29	1462	within	within	below	within
1/1/2026 4:28:09	14.54	11.88	17.71	1838	within	within	within	within
1/1/2026 7:43:17	29.67	65.79	13.4	114	within	within	below	within
1/1/2026 8:33:59	19.5	24	17.41	2080	within	within	within	above
1/1/2026 11:57:38	21.81	32.14	26.95	1176	within	within	above	within

Figure 1: Reading Status Table

2 The Dashboard

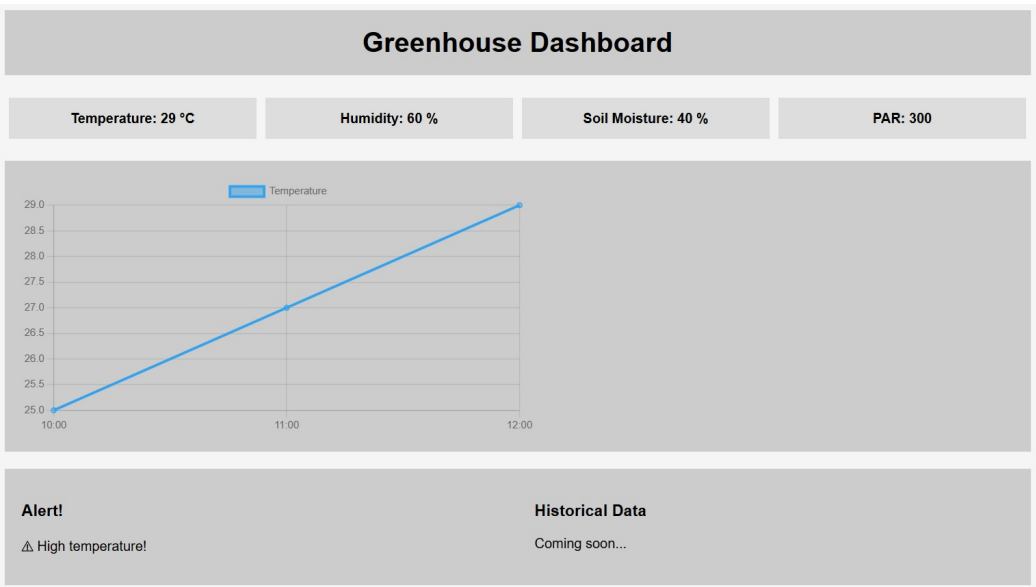


Figure 2: AI Generated HTML Template

Now that I have created the database, I started designing the website to show the greenhouse dashboard. I asked chatgpt to generate an HTML template for me based on my design which gave me series of code and the result is shown in Figure 2.

I was quite amazed at the result that AI gave me. Though it still needs a lot of work to really reflect the data from the database that I have created, it was a great help.

3 Backend - Frontend Development

I tried to integrate the data from the database to the HTML template but unfortunately, I could not seem to get it right. See Figure 3. There it does not show any data nor the graph at all. As I checked the HTML template, I realized there were issues with the labelling mismatch.

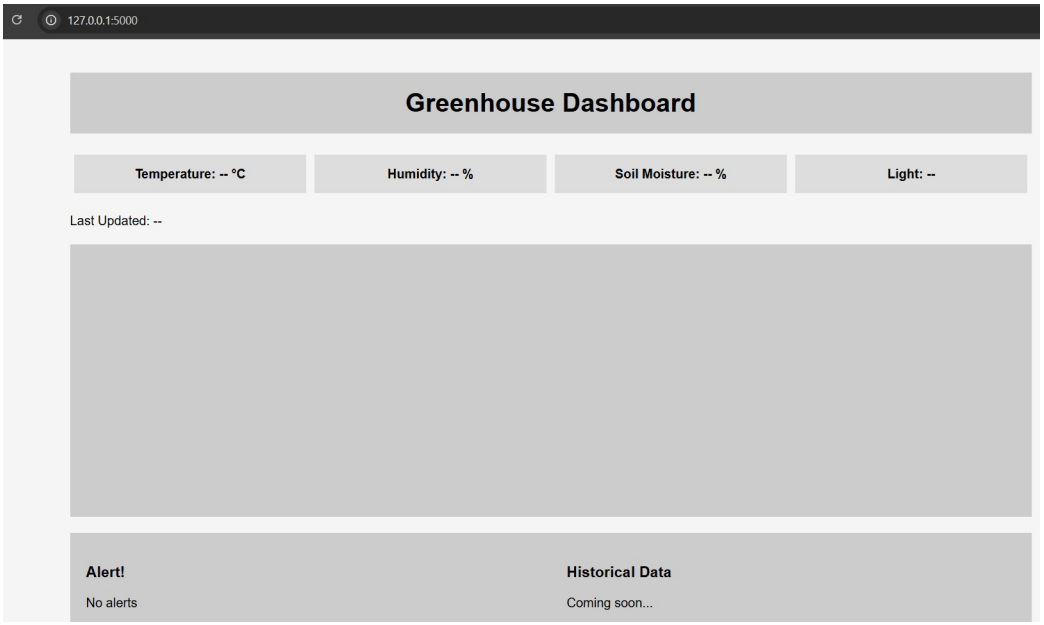


Figure 3: Dashboard Disconnection with the DB

I made sure that the labels and the columns matched between the HTML template and the database and it fixed the issue as shown in Figure 4. There it shows the last data entry from the database. However, the graph was showing the data in descending order. And it is showing only the temperature and does not switches between containers.

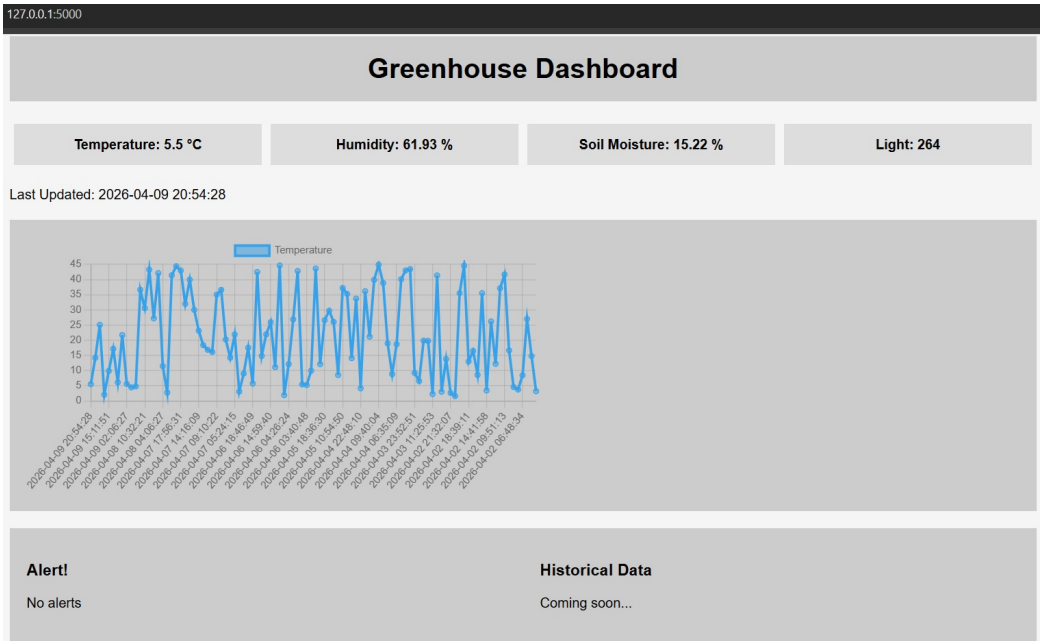


Figure 4: Dashboard and Database Connection Fixed

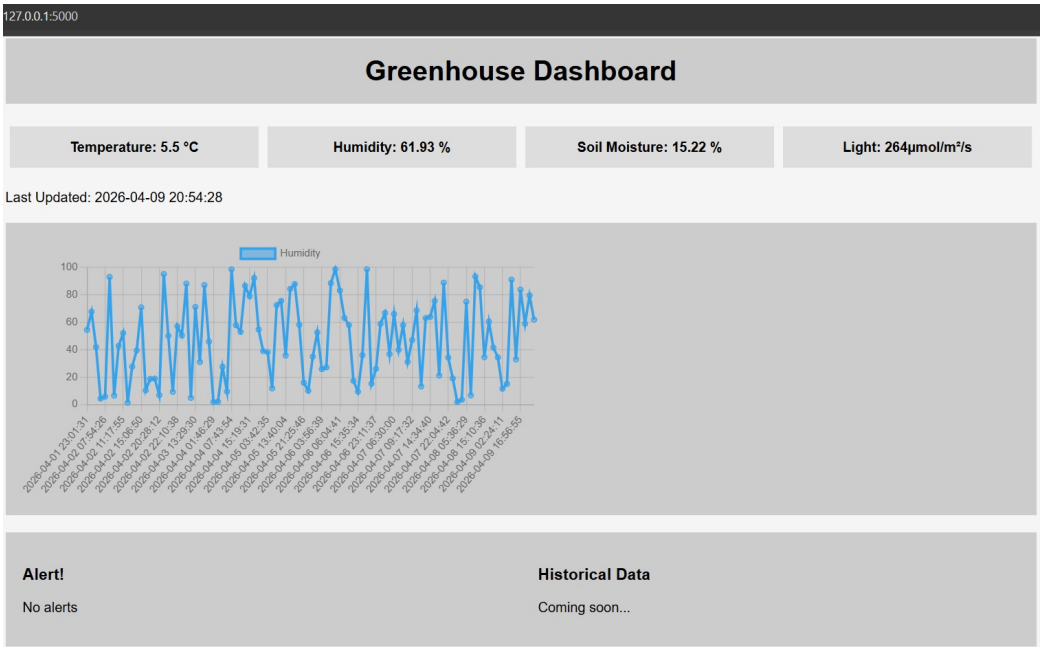
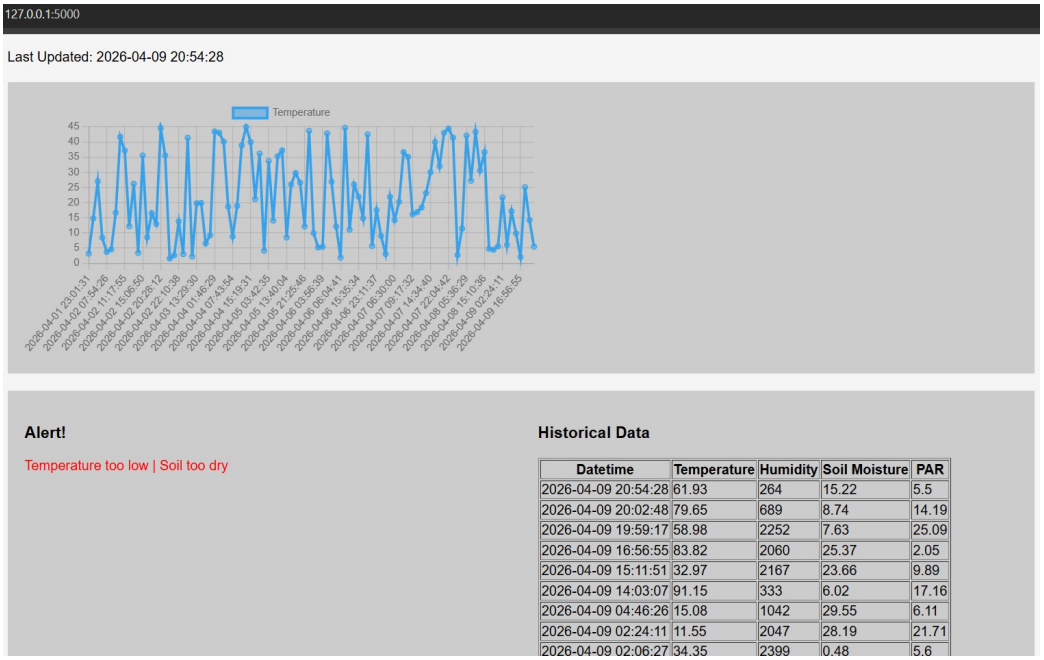


Figure 5: Graph Switches Data from Temperature to Light

I worked on updating the HTML template to make the dashboard more interactive which will allow me to click on containers between temperature, humidity, soil_moisture, and light. In Figure 5, I was able to make that happen. I can now switch and see the graph between temperature, humidity, soil_moisture, and the light. I also fixed the order of the data showing on the graph to show them in ascending order and added the unit for the PAR measurement "Light" container.

Now I need to work on the alerts and notification as well as the historical data where farmers can view the database. I did some research on how can that be done. I got some useful suggestion on how to do it with programming codes but I was having trouble connecting it to the database and kept getting the "No Alerts" result. I then realized that it was an issue with the position of the code in the HTML template. I did some rearrangements and was able to manage to incorporate the alerts to show on the dashboard as shown in Figure 6.



Datetime	Temperature	Humidity	Soil Moisture	PAR
2026-04-09 20:54:28	61.93	264	15.22	5.5
2026-04-09 20:02:48	79.65	689	8.74	14.19
2026-04-09 19:59:17	58.98	2252	7.63	25.09
2026-04-09 16:56:55	83.82	2060	25.37	2.05
2026-04-09 15:11:51	32.97	2167	23.66	9.89
2026-04-09 14:03:07	91.15	333	6.02	17.16
2026-04-09 04:46:26	15.08	1042	29.55	6.11
2026-04-09 02:24:11	11.55	2047	28.19	21.71
2026-04-09 02:06:27	34.35	2399	0.48	5.6

Figure 6: Alert and Historical Data View

The table under the historical data section was the right data I would like for the farmers to view, only that, I would like it to show as a different page and not on the main page for it does not look really pretty this way. And to do this, I need to add a new page in my Python file via flask by the following function:

```
@app.route("/")
```

In order to integrate the table to this new page, I created a new html template (table.html) for the table using the "buildTable()" function. This table must fetch from the database rendered in json format as illustrated in Figure 7.

```
  async function buildTable() {
try {
  const response = await fetch('/api/data');
  const data = await response.json();

  const table = document.createElement('table');
  table.border = "1";
```

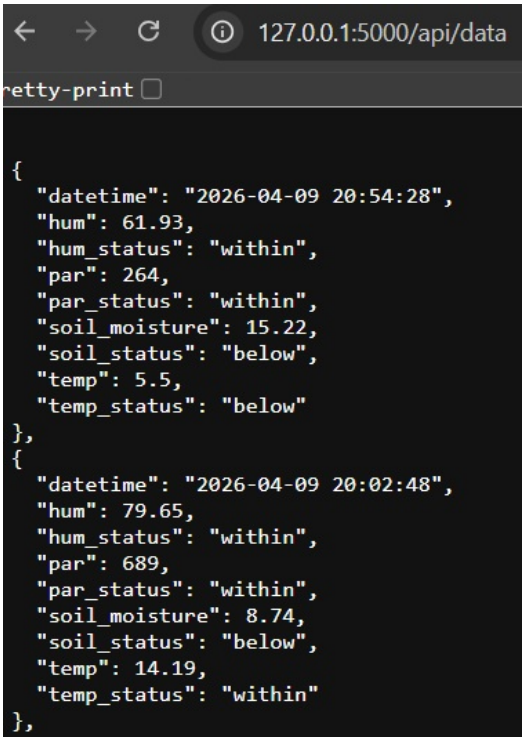


Figure 7: Database returned in JSON format

I tried opening the table.html template that I just created and it showed me an error as displayed in Figure 8. The template table is not picking up from the database.

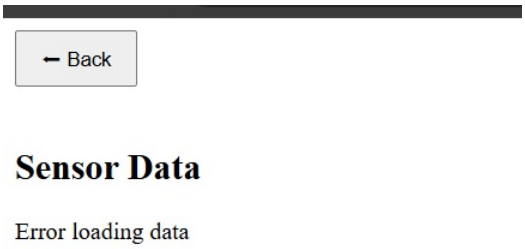


Figure 8: New Page Error

So I went back to my .py file and defined the new page as follows:

```
    @app.route("/api/historical")
def hist_data():
    db = get_db()
    rows = db.execute(f"SELECT * FROM readings_with_status ORDER BY datetime DES

    if not rows:
        return "No data found in table"

    columns = rows[0].keys()

    data = {col: [row[col] for row in rows] for col in columns}
    latest = rows[-1]

    return render_template(
        "table.html",
        data=data,
        columns=columns,
        latest=latest
    )
```

I then updated the division container in my main page html template as defined below:

```
    <div class="box">
            <h3>Historical Data</h3>
            <p id="historical">
                View Table

            </p>
        </div>

<script>
document.getElementById('historical').addEventListener('click', () => {
  window.location.href = '/api/historical';
});
</script>
```

After making sure that the data matched the html template and the .py file, I ran it and got an updated dashboard as seen in Figure 9. I like this view better because it looks neat and clean.

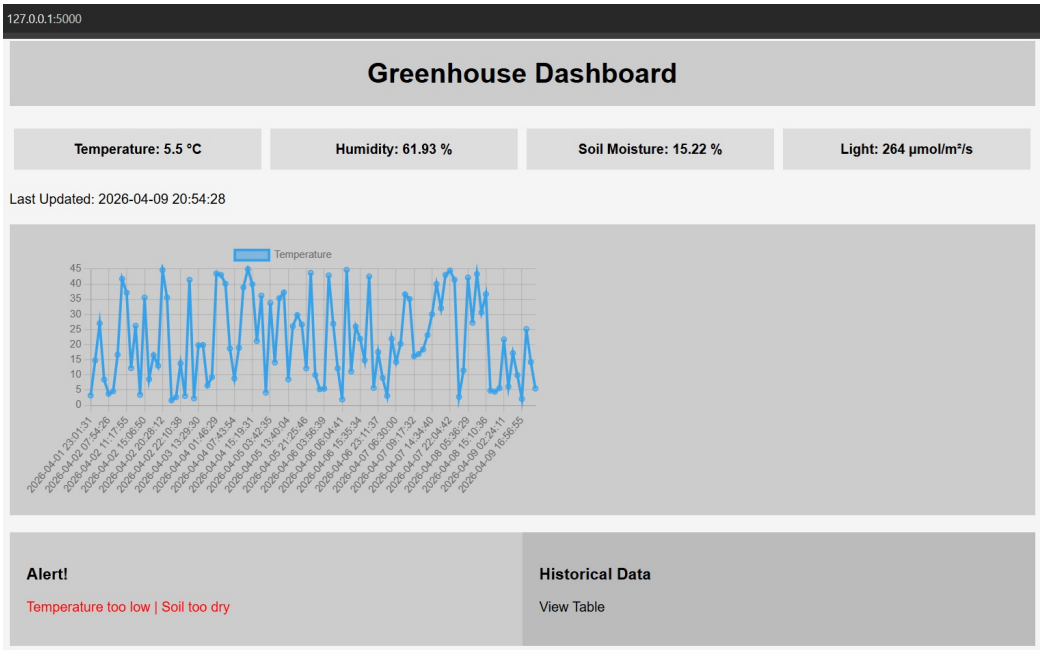
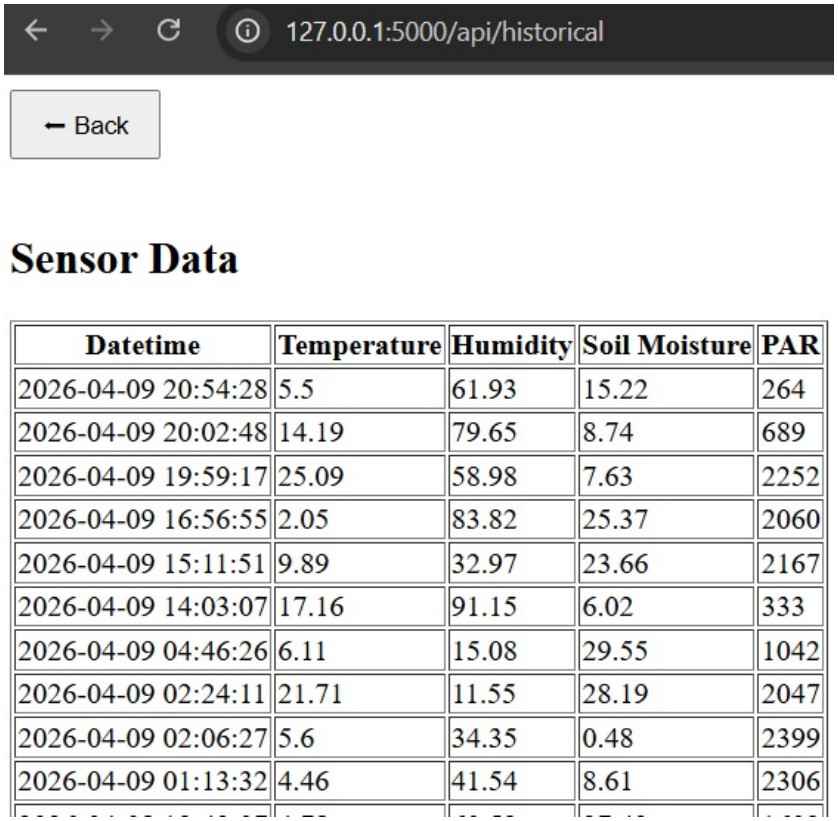


Figure 9: Updated Dashboard Main Page

Once the "View Table" button on this main page is clicked, it will open the second page that consists of the historical table, as illustrated in Figure 10. On this page, the user can view the historical data showing the latest data at the top and he may scroll down to see the rest.

This was basically the process I went through to realize this project. I am quite satisfied with it. However, I am open for feedback and improvements can still be applied during the validation part of this project. Until then.



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2026-04-09 16:56:55	2.05	83.82	25.37	2060
2026-04-09 15:11:51	9.89	32.97	23.66	2167
2026-04-09 14:03:07	17.16	91.15	6.02	333
2026-04-09 04:46:26	6.11	15.08	29.55	1042
2026-04-09 02:24:11	21.71	11.55	28.19	2047
2026-04-09 02:06:27	5.6	34.35	0.48	2399
2026-04-09 01:13:32	4.46	41.54	8.61	2306

Figure 10: Historical Data Page

4 Bibliography

References

[1] Mockaroo, LLC., (2026) https://mockaroo.com/

[2] Thompson, Grant L.; Kao-Kniffin, Jenny (2016), *Greenhouse environmental parameters.. PLOS ONE. Dataset.*, https://doi.org/10.1371/journal.pone.0155986.t003

[3] Gould, Scott, (15 May, 2024), *A Guide to Soil Moisture*, https://connectedcrops.ca/the-ultimate-guide-to-soil-moisture/

[4] Migro Grow Lights, (19 June 2024), *What is the best PAR intensity for plants? PPFD DLI adding CO2*, https://migrolight.com/blogs/grow-light-news/what-is-the-best-par-intensity-for-plants-optimum-ppfd-and-dli